

PPA-3: Polyhedral Boolean Encoding System

Title

Relational Boolean Encoding Using Polyhedral State Spaces

Field of the Invention

The invention relates to information encoding, Boolean logic representation, and computational state systems, particularly to encoding Boolean information using spatially symmetric polyhedral structures rather than Cartesian bit axes.

Background

Traditional Boolean systems encode bits as independent, axis-aligned values (0/1). This representation:

- destroys chirality,
- privileges coordinate systems,
- and fails to preserve relational structure under rotation or transformation.

There is a need for a Boolean encoding system that:

- preserves symmetry,
 - embeds bits relationally,
 - and remains Boolean-complete.
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Summary

The invention discloses a **polyhedral Boolean encoding system** in which Boolean-complete state spaces are represented as symmetric polyhedra.

In a preferred embodiment:

- a tetrahedron encodes four states equivalent to two Boolean bits,
- where bits represent relational properties (e.g., spin, chirality),
- not orthogonal axes.

Higher-order polyhedra encode larger Boolean-complete state spaces.

Detailed Description

Tetrahedral Boolean Encoding

A tetrahedron has four vertices corresponding to four states:

$\{00, 01, 10, 11\}$

The bits are:

- relational,
 - symmetry-preserving,
 - invariant under allowed transformations.
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Non-Cartesian Bit Representation

Bits are not mapped to X/Y/Z axes.

They are defined by:

- orientation,
 - handedness,
 - or other symmetry-invariant properties.
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Extension to Higher Polyhedra

The encoding generalizes to:

- cubes,
 - dodecahedra,
 - icosahedra,
 - and composite polyhedral systems.
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Advantages

- Boolean-complete

- Preserves chirality
- Rotation-invariant
- Compatible with physical systems
- Avoids coordinate bias

Provisional Statement

This application claims priority for relational Boolean encoding systems using polyhedral state spaces.